QUADS

Economic and Social Data Service

Smart Qualitative Data: Methods and Community Tools for Data Mark-Up SQUAD

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New qualitative data UK initiative

- Demonstrator Scheme for Qualitative Data Sharing and Research Archiving scheme - QUADS

- main aim of scheme to develop and promote innovative methodological approaches to the archiving, sharing, re-use and secondary analysis of qualitative research and data
  - models may be of temporary, local or thematic archiving
  - complement the ESDS Qualidata approach (traditional data archiving model)
  - exploit new or existing research collaborations locally, nationally or internationally

- explore a range of new models for increasing access to qualitative data resources, and for extending the reach and impact of qualitative studies

- draw primarily on existing qualitative research and data sets of a range of types but encourages researchers to explore the use of stored and shared video, visual and audio data sets

- promote understanding of the benefits and challenges of emerging information and communication e-science technologies

- aim to disseminate good practice in qualitative data sharing and research archiving

- part of the ESRC's initiative to increase the UK resource of highly skilled researchers, and to fully exploit the distinctive potential offered by qualitative research and data

- @£500,000 over 10 months: 6 awards – 5 demonstrators + 1 coordination

SQUAD Aims

- collaboration between UK Data Archive, University of Essex and Language Technology Group, Human Communication Research Centre, School of Informatics, University of Edinburgh
- Essex lead partner
- 18 months duration, 1 March 2005 – 31 august 2006
- 5 part-time staff split across sites = 1 FTE

Aims:

- to explore methodological and technical solutions for 'exposing' digital qualitative data to make them fully shareable and exploitable and to promote appropriate standards and tools
- Precursors of data sharing and collaborative research practice and data analysis are to found in the methods and tools for documenting and representing data

Why do we need tools & standards?

- to archive and web-enable high quality qualitative data in a way that faithfully represents its origins and context
- to provide rich and full documentation that enables effective resource discovery (already do DDI first 3 levels)
- to enable creative and exciting ways of exploring and visualizing data
  - from simple publishing of anonymised digital qualitative data
  - through mark-up to the ability to link qualitative data to other distributed data sources (e.g. audio-visual or geo-coded data sources)
- the absence of appropriate tools and standards is inhibiting successful digitisation efforts
  - many popular qualitative collections are not yet even in digital format
  - "digitising" these collections is often merely providing an online catalogue of metadata
  - there is little community knowledge in this area about the use of standards (TEI not used in social science)

Prerequisites for making data shareable

- data are collected to a high standard
- research methods and practices (including consent process) are fully documented
- the context of the data collection and analysis is captured
- the richness of the structure and features of data and are made available (use of mark-up)
- the interrelationships between data and analyses (intra-project) are made available (issues of representation)
- data are represented in intuitive, appealing and sensitive ways that satisfy the ethical and legal requirements to which they are bound

Main objectives

- specify, test and propose an XML schema for storing and marking-up a broad range of qualitative data types
  - textual or audio-visual social science data
  - and for e-social science exploitations, i.e. grid-enabling data
  - ESDS Qualidata had developed draft DTD based on TEI)
- investigate requirements for contextualising data (e.g. interview setting and interviewer characteristics), and develop standards for data documentation and common vocabularies
- develop user-friendly (java-based) tools for semi-automating processes (using NLP technologies) already used to prepare qualitative data for digital archiving and e-science type exploitation
- investigate non-proprietary tools for publishing and archiving XML marked-up data and study context - Qualitative Data Mark-up Tools (QDMT). Enable preservation of data structures and links to other objects
- increase awareness and provide training with step-by-step guides and exemplars on the use of these tools and standards utilised

A uniform quali format

- a **uniform format** for richly encoding qualitative research is necessary as it:
  - ensures consistency across datasets
  - supports the development of common web-based publishing and search tools
  - and facilitates data interchange and comparison among datasets

- it could also enable data and linked products to be imported and exported directly into and out of CAQDAS packages, avoiding the reliance on just a single product, and offering the opportunity to share analytic workings outside the confines of the particular software

- a draft but limited formal definition of a common XML vocabulary and Document Type Definition (DTD) based on the Text Encoding Initiative (TEI) for describing these structures has been prepared by ESDS Qualidata

- but the important development of a common framework for marking up the content of qualitative datasets **requires support and contribution** from various sectors of the social science community:

  - data creators
  - qualitative data software developers
  - data archivists
  - end users

- fortunately, the expansion of e-science funding is accelerating the need for such standards – exposure of 'structured' qualitative data to the web.

Marking up what?

- spoken interview texts provide the clearest -and most common -example of the kinds of encoding features needed
- three basic groups of features
  - structural features representing basic format: utterance, specific turn taker, other speech tags e.g. defining idiosyncrasies
  - structural features representing links to other data types created in the course of the research process (e.g. audio or video referencing points, researcher annotations)
  - structural features representing identifying information such as real names, company names, place names, temporal information

Solutions to qualitative data mark-up with XML: Qualitative Data Mark-up Tools (QDMT)

- **systematic preparation of digital data** : to create formatted text documents ready for xml output

- **mark-up of data to capture basic structural features of textual data**: e.g. turn-takers, speakers and selected demographic details

- **advanced annotation or mark-up of data**

  - **automated information extraction of basic semantic information:** inserting tags for real names and temporal references
  - **automated anonymisation:** replacing names with dummy forms, including co-references
  - **geographic mark-up to enable data linking:** identifying and applying geographic mark-up, and scoping researchers' needs for geo-linking

- **basic classification or thematic coding of textual data:** for of efficient resource discovery rather than data analysis; will investigate linking into a domain ontology (e.g. social science thesaurus) - Key word assignment tool

- **contextual documentation to capture richness of the research methods, data collection and analytic interpretation and representation:** will dovetail with Cardiff QUADS project to look at the interrelationships between complex intra-project data, annotations and context

- **exposure of annotated and contextualised qualitative data to the web:** investigating publishing of above QDM XML outputs to ESDS Qualidata Online, opportunities for exchange within CAQDAS tools, etc.

First output from automated mark-up

Z:\Qualidata\OpenAccess\zip\int010-out-nadeem.html - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Address Z:\Qualidata\OpenAccess\zip\int010-out-nadeem.html

Key: organisation person location date time money percent

sex: M/FM
dob: 1921
Place: Oldham
finalocc: Oldham

{START LOUISE BECKWITH 1A}

U id='1' who='interviewer' Can you tell me your name?
U id='2' who='subject' Louise Margaret Beckwith.
U id='3' who='interviewer' And your address?
U id='4' who='subject' {Missing}, Bedlington Station.
U id='5' who='interviewer' And your date of birth?
U id='6' who='subject' Twenty-first of 6th '70.
U id='7' who='interviewer' So you are 18.
U id='8' who='subject' Yes.
U id='9' who='interviewer' And where were you born?
U id='10' who='subject' Ashington Hospital.
U id='11' who='interviewer' And you were born and lived from this house?
U id='12' who='subject' I've always lived here.
U id='13' who='interviewer' I'd like to ask you about your relationship with your grandparents. Are they still alive?
U id='14' who='subject' Me grandma.
U id='15' who='interviewer' That's on your mum's side?
U id='16' who='subject' Mum's side.
U id='17' who='interviewer' What about on your dad's side?
U id='18' who='subject' Neither. Me granddad died when me dad was 14. And me nana died just after mam's dad.
U id='19' who='interviewer' So you can remember your nana, your dad's mum?
U id='20' who='subject' Yeah.
U id='21' who='interviewer' How often do you see your grandma?
U id='22' who='subject' Nearly every day. I see her - if I don't see her during the week I see her Friday, Saturday, Sunday. So - sometimes more than once a day.
U id='23' who='interviewer' Did you used to see your mum's mum more than you saw your nana?
U id='24' who='subject' Yes.

Existing tools

- Making use of unix-based community tools used in NLP fields
- applications are for mining and summarising e.g. legal, pharmaceutical reports, news stories, web sites etc.
- but not tested on for social science corpora yet – training data is limited
- tools using named entity recognition and speech taggers will insert xml tags
- others use stand-of annotation (x-link, x-pointer etc)
- Currently unfriendly tools - need GUIs!

Relationship to ESDS Qualidata

- ESDS Qualidata, through the UKDA, currently provides the ESRC RRB strategy for archiving, accessing and supporting users of qualitative research data

- strong emphasis on
  - developing community standards for describing data/metadata
  - providing better study and data context to inform re-use

- grant represents critical useful R&D funding for ESDS Qualidata who have no budget to do this normally

- SQUAD outputs and tools will be used for in-house processing of qualitative data

- and made available as shareable standards and tools for others archiving data

Summary of deliverables I

– report on consultation with, and initial assessment by, LTG at Edinburgh, and a consolidated plan of work	Month 2
– report on applying levels of mark-up, setting out minimal and ideal requirements for different data types (interview data, field notes, naturally occurring speech, etc.)	Month 5
– report on first set of components of the Qualitative Data Mark-up suite of tools, including user testing results	Month 9
– report on second batch of components of the Qualitative Data Mark-up suite of tools, including user testing and user workshop	Month 15
– short promotional overview of QDM tools and applications	Month 15

Summary of deliverables II

– draft user guide and tutorials for each data preparation process and tool, with exemplars	Month 16
– tool and programming documentation	Month 16
– report on further needs and developments for components that may not be completed	Month 17
– report on fit of tools to ESDS Qualidata Online system	Month 17
– report of brief evaluation of user guide and tutorials	Month 17
– final report	Month 18